

SHRAVAN KANNEPALLI

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Education

University of Maryland, College Park

Bachelor of Science in Mechanical Engineering

- 2024 & 2025 Fall Semester Dean's List

May 2028

College Park, MD

Work Experience

NSF Sensor Development and Implementation Pipeline (SDIP) Undergraduate Research

June 2026 – August 2026

Clarkson University

Potsdam, NY

- Integrated a SPS30 air quality sensor onto a UAV/drone via a custom designed bracket & Arduino-based data logging unit.
- Executed standardized flight paths and conducted wind tunnel testing at velocities up to 20 m/s, alongside controlled flight tests, to validate structural integrity and aerodynamic stability.
- Performed graphical and statistical analyses across a ground propeller system to establish quantitative correlations between drone downwash and SPS30 sensor readings.

Geospatial Analysis Intern

June 2023 – August 2023

NASA

Greenbelt, MD

- Engineered a geospatial database by scraping and analyzing AWS satellite data, enabling the processing of complex queries to monitor global environmental shifts.
- Developed a real-time tracking system for significant Earth events, providing the scientific community with automated updates and high-fidelity data insights.

Projects

L1 Rocket Certification

October 2025 – November 2025

- Earned L1 Rocket Certification by designing and successfully launching a high-power model rocket within an accelerated development cycle.
- Utilized OpenRocket simulation software to model complex flight dynamics and center of pressure/gravity relationships to validate aerodynamic stability.
- Engineered recovery systems and verified flight profiles through digital simulations to ensure safe vehicle descent and hardware reuse.

Over-Terrain Autonomous Vehicle

September 2024 – December 2024

- Collaborated within a five-person team to design an autonomous Over-Terrain Vehicle, earning a Top 5 Innovative Project Nomination.
- Integrated a custom collection system featuring a puck-and-aligner mechanism, pump, limit switch, and conductivity sensor, interfaced via Arduino to automate water sampling.
- Served as Budget Analyst for raw materials and electrical components to ensure the project remained within strict financial constraints.
- Contributed to vehicle construction, assisting in the physical assembly and structural optimization of the chassis to navigate varied terrain.

Activities & Leadership

Terrapin Rocket Club | Aerostructures / Two Stage

College Park, MD

- Collaborate with team members on design, fabrication, and integration challenges to support the development and successful launch of competition rockets.
- Worked on classifying failure analysis and building of the main/drogue parachute for next year's competition rocket.

FIRST LEGO League (FLL) Mentoring | Mentor

Regional

- Mentored a FIRST LEGO League (FLL) team by guiding students through the robot design cycle and a specialized research project.
- Facilitated technical growth by teaching programming logic, iterative testing, and fundamental engineering principles.
- Instilled core design values and professional ethics to help the team translate creative concepts into functional, competition-ready solutions.

DaCadence UMD Acappella | Vocal Percussion

College Park, MD

- Work with musical directors to notate drum parts for arrangement
- Lead voluntary sessions to teach vocal percussion skills to high school and college acapella groups.

Carnatic Vocalist | Performer

Chennai, India & USA

- Part of a vocal duo performing south indian classical (carnatic) music concerts
- Deliver regular live performances for cultural organizations and regional venues across Maryland and the Washington, D.C. area.
- Perform in the yearly Chennai music festival in December at different Sabhas

Technical Skills

CAD & Hardware: SolidWorks, Fusion 360, Circuit Design, Soldering

Coding Languages: Python, Java, C++, MATLAB, Arduino

Tools & Methods: OpenRocket, Data Logging, Aerodynamic Analysis

Soft Skills: Mentoring, Technical Teaching, Budget Analysis